Efficacy and Safety of Lipidara in Heterozygous familial hypercholesterolaemia: A Phase 3 Randomised, Double-Blind, Placebo-Controlled Trial

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Note: Entirely fictional study with invented drug and data, created for training. Do not use for clinical decisions.

Abstract

Background: Lipidara (LPD) is a quarterly injectable siRNA therapy under investigation for heterozygous familial hypercholesterolaemia. **Methods:** In this multicentre trial, 388 adults were randomised 1:1 to Lipidara or placebo for 52 weeks, stratified by baseline severity and region. The primary endpoint was LDL-C percent change at day 90. **Results:** The primary endpoint was -51.0 with Lipidara versus -2.5 with placebo (p <0.001). Injection-site pain occurred in 7.9% of Lipidara patients and 3.1% of placebo patients. **Conclusions:** Lipidara met its primary endpoint with a manageable safety profile in heterozygous familial hypercholesterolaemia.

Methods

Eligible participants were aged 18 to 75 years with a confirmed diagnosis of heterozygous familial hypercholesterolaemia and provided written informed consent. Exclusion criteria included pregnancy, severe hepatic impairment (Child-Pugh C), and concurrent use of strong CYP3A4 inducers. Randomisation used an interactive web-response system. The analysis followed the intention-to-treat principle with multiple imputation for missing data.

Results

Characteristic	Lipidara	Placebo
Participants (n)	194	194
Mean age (years)	52.6	56.5
Female (%)	42	56
Primary endpoint	-51.0	-2.5
p-value	<0.001	-
Injection-site pain (%)	7.9	3.1
Discontinuation due to AE (%)	4.3	3.5

Serious adverse events were reported in 8.5% of the Lipidara group and 8.7% of the placebo group. There were 5 deaths in the Lipidara arm and 3 in the placebo arm, none judged related to study treatment.

Discussion

The effect size observed with Lipidara is clinically meaningful. Limitations include the 52-week duration, under-representation of patients aged over 75 years, and limited ethnic diversity. Long-term safety beyond one year requires an open-label extension, which is ongoing.

References

1. Synthetic reference list omitted. 2. Illustrative Guidelines on Management, 2025.